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**Activity 6: Targeting Specific Nodes and Managing Services**

1. **Objectives:**

1.1 Individualize hosts
1.2 Apply tags in selecting plays to run
1.3 Managing Services from remote servers using playbooks
1.4 Managing Ansible Roles for Modularization

2. **Intended Learning Outcomes:**

- Differentiate and organize remote hosts into appropriate inventory groups based on their server roles (e.g., web, database, file servers).
- Analyze playbook execution results and interpret changes made across different remote servers.
- Configure and manage system services remotely using Ansible modules (e.g., service module for starting and enabling services).
- Design and apply Ansible Roles to modularize configurations and improve playbook reusability, scalability, and maintainability.

3. **Discussion**:

In this activity, we try to individualize hosts. For example, we don't want apache on all our servers, or maybe only one of our servers is a web server, or maybe we have different servers like database or file servers running different things on different categories of servers and that is what we are going to take a look at in this activity.

We also try to manage services that do not automatically run using the automations in playbook. For example, when we install web servers or httpd for CentOS, we notice that the service did not start automatically.

**Requirement:**

In this activity, you will need to create another Ubuntu VM and name it Server 3. Likewise, you need to activate the second adapter to a host-only adapter after the installations. Take note of the IP address of the Server 3. Make sure to use the command *ssh-copy-id* to copy the public key to Server 3. Verify if you can successfully SSH to Server 3.

**Task 1: Targeting Specific Nodes**

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.

```yaml
- hosts: all
  become: true
  pre_tasks:


   - name: install update (CentOS)
     dnf:
      update_only: yes
      update_cache: yes
     when: ansible_distribution == "CentOS"

   - name: install update (CentOS)
     apt:
      updgrade: yes
      update_cache: yes
     when: ansible_distribution == "Ubuntu"
```

2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

```
inventory.ini ×   site.yml ●   ansible.cfg

inventory.ini
[isaac]
192.168.56.114

[jhon]
192.168.56.115

[alegre]

192.168.56.117
```

Make sure to save the file and exit.

Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.

3. Edit the *site.yml* by following the image below:

```yaml


- hosts: all
  become: true
  pre_tasks:


  - name: install update (CentOS)
    dnf:
     update_only: yes
     update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install update (CentOS)
    apt:
     updgrade: yes
     update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu Servers
    apt:
     name:
       - apache2
       - libapache2-mod-php
     state: latest
     update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS Servers
    dnf:
     name:
```

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Make sure to save the file and exit.

The *pre-tasks* command tells the ansible to run it before any other thing. In the *pre-tasks*, CentOS will install updates while Ubuntu will upgrade its distribution

package. This will run before running the second play, which is targeted at *web_servers*. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.

Run the *site.yml* file and describe the result.

4. Let's try to edit again the *site.yml* file. This time, we are going to add plays targeting the other servers. This time we target the *db_servers* by adding it on the current *site.yml*. Below is an example: (Note add this at the end of the playbooks from task 1.3.

```yaml
- hosts: db_servers
  become: true
  tasks:

    - name: install mariadb package(CentOS)
      yum:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "CentOS"

    - name: install mariadb package(Ubuntu)
      apt:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: "Mariadb- Restarting/Enabling"
      service:
        name: mariadb
        state: restarted
        enabled: true
```

Make sure to save the file and exit.

Run the *site.yml* file and describe the result.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: *systemctl status mariadb.* Do this on the CentOS server also.

Describe the output.

6. Edit the *site.yml* again. This time we will append the code to configure installation on the *file_servers* group. We can add the following on our file.

```
- hosts: file_servers
  become: true
  tasks:

    - name: install samba package
      package:
        name: samba
        state: latest
```

Make sure to save the file and exit.

Run the *site.yml* file and describe the result.

The testing of the *file_servers* is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

**Task 2: Using Tags in running playbooks**

In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the *site.yml* file. Add tags to the playbook. After the name, we can place the tags: *name_of_tag*. This is an arbitrary command, which means you can use any name for a tag.

```yaml
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    tags: always
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

```yaml
- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    tags: apache,apache2,ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    tags: apache,centos,httpd
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    dnf:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb packege (Ubuntu)
    tags: db, mariadb,ubuntu
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"

- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    tags: samba
    package:
      name: samba
      state: latest
```

Make sure to save the file and exit.

Run the *site.yml* file and describe the result.

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook site.yml -K

PLAY [file_servers] ***************************************************************************************
***************

TASK [Gathering Facts] ************************************************************************************
***************
ok: [192.168.56.117]

TASK [install samba package] ******************************************************************************
***************
changed: [192.168.56.117]

PLAY RECAP ************************************************************************************************
***************
192.168.56.114             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=
192.168.56.115             : ok=5    changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=
192.168.56.117             : ok=4    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=
```

2. On the local machine, try to issue the following commands and describe each result:

   *2.1 ansible-playbook --list-tags site.yml*

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook --list-tags site.ym

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: [apache2,apache,ubuntu]
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

## 2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook --tags centos --ask-become-pass site.yml
[WARNING]: Platform linux on host 192.168.56.115 is using the discovered Python interpreter at /usr/bin/python3
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information
ok: [192.168.56.115]
[WARNING]: Platform linux on host 192.168.56.117 is using the discovered Python interpreter at /usr/bin/python3
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information
ok: [192.168.56.117]
[WARNING]: Platform linux on host 192.168.56.114 is using the discovered Python interpreter at /usr/bin/python3
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information
ok: [192.168.56.114]

PLAY [web_servers] ************************************************************************************

TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.114]

PLAY [db_servers] *************************************************************************************

TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.115]

PLAY [file_servers] ***********************************************************************************

TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.117]

PLAY RECAP ********************************************************************************************
192.168.56.114             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignore
192.168.56.115             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignore
192.168.56.117             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignore
```

## 2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook --tags db --ask-become-pass site.yml
TASK [install update (Ubuntu)] ************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.117]
ok: [192.168.56.115]

PLAY [web_servers] ************************************************************************************

PLAY [db_servers] *************************************************************************************

TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.115]

TASK [install mariadb package(CentOS)] ****************************************************************
skipping: [192.168.56.115]

TASK [install mariadb package(Ubuntu)] ****************************************************************
ok: [192.168.56.115]

PLAY [file_servers] ***********************************************************************************

TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.117]

PLAY RECAP ********************************************************************************************
192.168.56.114             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.115             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.117             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

*2.4 ansible-playbook --tags apache --ask-become-pass site.yml*

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook --tags apache --ask-become-pass site.yml
ok: [192.168.56.117]
ok: [192.168.56.115]
ok: [192.168.56.114]

PLAY [web_servers] *************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.114]

TASK [install apache and php for Ubuntu Servers] *******************************************************
ok: [192.168.56.114]

TASK [install apache and php for CentOS Servers] *******************************************************
skipping: [192.168.56.114]

PLAY [db_servers] **************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.115]

PLAY [file_servers] ************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.117]

PLAY RECAP *********************************************************************************************
192.168.56.114             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.115             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.117             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.5 *ansible-playbook --tags “apache,db” --ask-become-pass site.yml*

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook --tags "apache,db" site.yml -K
TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.114]

TASK [install apache and php for Ubuntu Servers] *******************************************************
ok: [192.168.56.114]

TASK [install apache and php for CentOS Servers] *******************************************************
skipping: [192.168.56.114]

PLAY [db_servers] **************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.115]

TASK [install mariadb package(CentOS)] *****************************************************************
skipping: [192.168.56.115]

TASK [install mariadb package(Ubuntu)] *****************************************************************
ok: [192.168.56.115]

PLAY [file_servers] ************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.56.117]

PLAY RECAP *********************************************************************************************
192.168.56.114             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.115             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.117             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

**Task 3: Managing Services**

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

```
- name: install apache and php for CentOS servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos,httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

Figure 3.1.1
Make sure to save the file and exit.

You would also notice from our previous activity that we already created a module that runs a service.

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    dnf:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true
```

Figure 3.1.2

This is because in CentOS, installed packages' services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command *sudo systemctl stop httpd*.

When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

3. Go to the local machine and this time, run the *site.yml* file. Then after running the file, go again to the CentOS server and enter its IP address on the browser. Describe the result.

   To automatically enable the service every time we run the playbook, use the command *enabled: true* similar to Figure 7.1.2 and save the playbook.

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook site.yml -K
PLAY [db_servers] ***********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [192.168.56.115]

TASK [install mariadb package(CentOS)] **************************************************************************
skipping: [192.168.56.115]

TASK [Mariadb- Restarting/Enabling] *****************************************************************************
changed: [192.168.56.115]

TASK [install mariadb package(Ubuntu)] **************************************************************************
ok: [192.168.56.115]

TASK [Mariadb- Restarting/Enabling] *****************************************************************************
changed: [192.168.56.115]

PLAY [file_servers] *********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [192.168.56.117]

TASK [install samba package] ************************************************************************************
ok: [192.168.56.117]

PLAY RECAP ******************************************************************************************************
192.168.56.114             : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.115             : ok=6    changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.117             : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

isaac@Workstation:~/CPE212_Alegre/HOA6.1$
```

**Supplementary Activity:**

1. **Creating a Role for Web Servers**

site.yml | main.yml DB_Servers/... | main.yml Files_Servers/...

Web_Servers > tasks > main.yml

```yaml
---
# tasks file for Web_Servers

    - name: install apache and php for Ubuntu Servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS Servers
      tags: apache,centos,httpd
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

    - name: start httpd (CentOS)
      tags: apache, centos, httpd
      service:
        name: httpd
        stage: started
      when: ansible_distribution == "CentOS"
```

2. **Creating a Role for Database Servers**

DB_Servers > tasks > ! main.yml

```yaml
---
# tasks file for DB_Servers

- name: install mariadb package(CentOS)
  tags: centos, db,mariadb
  yum:
      name: mariadb-server
      state: latest
      when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/Enabling"
  service:
      name: mariadb
      state: restarted
      enabled: true


- name: install mariadb package(Ubuntu)
  tags: db, mariadb, ubuntu
  apt:
      name: mariadb-server
      state: latest
      when: ansible_distribution == "Ubuntu"

- name: "Mariadb- Restarting/Enabling"
  service:
      name: mariadb
      state: restarted
      enabled: true
```

3. **Creating a Role for Files Servers**

```
site.yml    main.yml DB_Servers/...    main.yml Files_Servers/.
Files_Servers > tasks > main.yml
---
# tasks file for Files_Servers

- name: install samba package
  tags: samba
  package:
    name: samba
    state: latest
```

4. **Combining Multiple Roles in a Playbook**

```
allinone.yml
- hosts: all
  become: true

  roles:
    - DB_Servers
    - Files_Servers
    - Web_Servers
```

**To orchestrate multiple roles in a single playbook for different server groups. Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for database.**

```
isaac@Workstation:~/CPE212_Alegre/HOA6.1$ ansible-playbook allinone.yml -K
BECOME password:

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.114 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.114]

TASK [Web_Servers : install apache and php for Ubuntu Servers] *************************
ok: [192.168.56.114]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.115 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.115]

TASK [DB_Servers : install mariadb package(Ubuntu)] ************************************
ok: [192.168.56.115]

TASK [DB_Servers : Mariadb- Restarting/Enabling] ***************************************
changed: [192.168.56.115]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.117 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
```

```
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.117]

TASK [Files_Servers : install samba package] *******************************************
ok: [192.168.56.117]

PLAY RECAP *********************************************************************
192.168.56.114             : ok=2    changed=0    unreachable=0    failed=0    skipped=0
    rescued=0    ignored=0
192.168.56.115             : ok=3    changed=1    unreachable=0    failed=0    skipped=0
    rescued=0    ignored=0
192.168.56.117             : ok=2    changed=0    unreachable=0    failed=0    skipped=0
    rescued=0    ignored=0

isaac@Workstation:~/CPE212_Alegre/HOA6.1$
```

**Reflections:**

Answer the following:

1. What is the importance of putting our remote servers into groups?
   - Grouping remote servers simplifies management by allowing administrators to apply configurations, updates, and monitoring collectively, ensuring consistency, scalability, and reduced overhead in large distributed environments.

2. What is the importance of tags in playbooks?
   - Tags in Ansible playbooks provide granular control, enabling selective execution of specific tasks or sections without running the entire playbook, which improves efficiency, flexibility, and troubleshooting.

3. Why do think some services need to be managed automatically in playbooks?
   - Automating service management in playbooks ensures reliability and consistency by handling repetitive tasks, enforcing desired states, and reducing human error, which is critical for maintaining smooth operations in complex systems.